

Tema 2

▪ Ejercicio 1. Sea D el dominio de

$$f(x, y) = \frac{\sqrt{2y - y^2 - x^2}}{\sqrt{y - x}}.$$

a) Grafique separadamente D , su frontera y su interior.

b) Grafique el conjunto de nivel 0. Indique cuáles de los siguientes puntos pertenecen a ese conjunto: $(1, 1)$, $(0, 2)$, $(0, 0)$.

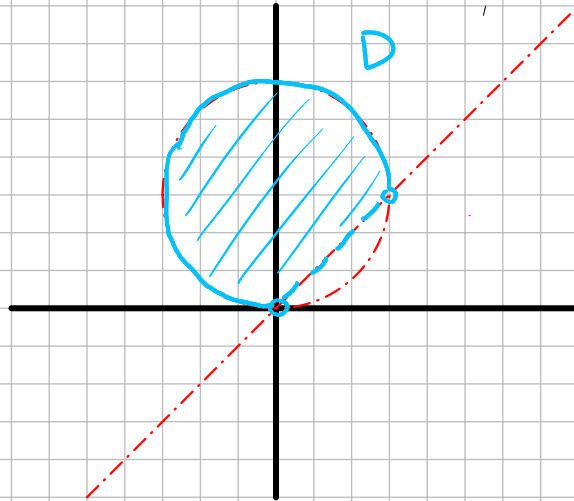
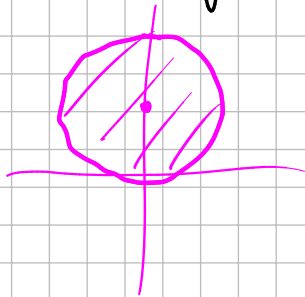
$$2y - y^2 - x^2 \geq 0 \quad \wedge \quad y - x > 0$$

$$x^2 + y^2 - 2y \leq 0$$

$$x^2 + y^2 - 2y + 1 - 1 \leq 0$$

$$x^2 + (y - 1)^2 \leq 1$$

$$y = x$$



∂D

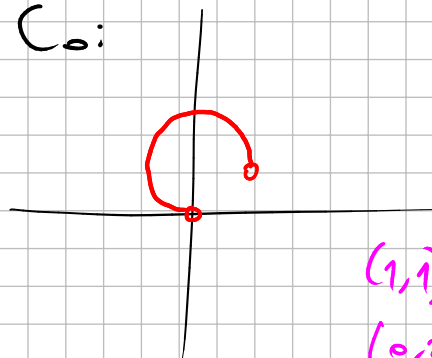
$\text{int } D$



$$0 = \frac{\sqrt{2y - y^2 - x^2}}{\sqrt{y - x}} \Rightarrow 2y - y^2 - x^2 = 0$$

$$x^2 + (y - 1)^2 = 1$$

C_0 :



SÓLO GRAFICAR
LO QUE PERTENECE
A D!!

$(1, 1)$ No
 $(0, 2)$ Si.
 $(0, 0)$ No.

■ Ejercicio 2. Halle los puntos críticos de $f(x,y) = y^3 - \frac{1}{2}xy^2 + \frac{1}{2}x$ y clasifíquelos.

$$\nabla f = \left(-\frac{1}{2}y^2 + \frac{1}{2}, 3y^2 - xy\right) = (0,0)$$

$$-\frac{1}{2}y^2 + \frac{1}{2} = 0$$
$$y^2 = 1 \rightarrow \begin{matrix} y=1 \\ y=-1 \end{matrix}$$

$$y=1$$

$$3-x=0$$
$$x=3$$

$$P.C.: (3,1), (-3,-1)$$

$$y=-1$$

$$3+x=0$$
$$x=-3$$

NO!
(3,-1)
(-3,1)

$$\nabla f = \left(-\frac{1}{2}y^2 + \frac{1}{2}, 3y^2 - xy\right)$$

$$f''_{xx} = 0$$

$$f''_{xy} = -y$$

$$f''_{yx} = -y$$

$$f''_{yy} = 6y - x$$

$$Hf(3,1) = \begin{bmatrix} 0 & -1 \\ -1 & 3 \end{bmatrix}$$

$$H_1 = 0$$
$$H_2 = -1$$

$$Hf(-3,-1) = \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix}$$

AMBOS
NO SON
EXTREMOS

$$H_1 = 0$$
$$H_2 = -1$$

▪ Ejercicio 3. Pruebe que la ecuación

$$\sin(y) + x^2 z^3 - e^y z + y + 2x = 2$$

$$F(x, y, z) = \sin y + x^2 z^3 - e^y z + y + 2x - 2$$

define implícitamente una función $z = f(x, y)$ de clase C^1 en un entorno de $(x_0, y_0) = (1, 0)$ tal que $z_0 = f(x_0, y_0) > 0$. Halle todos los versores $\vec{v}$ para los cuales $f'((1, 0), \vec{v}) = 0$.

$$f'(A, \vec{v}) = \nabla f(A) \cdot \vec{v}$$

$\uparrow$ f dif en A.

$$f \in C^1 \Rightarrow f \text{ dif.}$$

$$(I) F(1, 0, z_0) = z_0^3 - z_0 = 0$$

$$z_0(z_0^2 - 1) = 0$$

$$\begin{aligned} & \rightarrow z_0 = 0 \\ & \rightarrow z_0 = 1 \\ & \rightarrow z_0 = -1 \end{aligned}$$

$$z_0 > 0$$

$$F(x, y, z) = \sin y + x^2 z^3 - e^y z + y + 2x - 2$$

$$(II) \nabla F = (2x \cdot z^3 + 2, \cos y - e^y z + 1, 3x^2 z^2 - e^y)$$

cont en $E(1, 0)$ ✓

$$(III) F'_z(1, 0, 1) = 2 \neq 0 \quad \checkmark$$

$$f'_x(1, 0) = -\frac{F'_x(1, 0, 1)}{F'_z(1, 0, 1)} = -\frac{4}{2} = -2$$

$$\nabla f(1, 0) = (-2, -\frac{1}{2})$$

$$f'_y(1, 0) = -\frac{F'_y(1, 0, 1)}{F'_z(1, 0, 1)} = -\frac{1}{2} = -\frac{1}{2}$$

$$\nabla f(1, 0) \cdot \vec{v} = 0$$

$$(-2, -\frac{1}{2}) \cdot \vec{v} = 0$$

$$\vec{v}_1 = \frac{(\frac{1}{2}, -2)}{\|(\frac{1}{2}, -2)\|} = \frac{(\frac{1}{2}, -2)}{\frac{\sqrt{17}}{2}} = (\frac{1}{\sqrt{17}}, -\frac{4}{\sqrt{17}})$$

$$\vec{v}_2 = (-\frac{1}{\sqrt{17}}, \frac{4}{\sqrt{17}})$$

No son versores!

$$\begin{aligned} & \left(\frac{1}{2}, -2\right) \\ & \rightarrow (-2, -\frac{1}{2}) \\ & \rightarrow (-\frac{1}{2}, 2) \end{aligned}$$

- Ejercicio 4. Sea $g(x, y) = f(xy, x^2 + y^2)$ con $f(r, s)$ de clase $C^1(\mathbb{R}^2)$ tal que $P(r, s) = 1 - r + s$ es su polinomio de Taylor de orden 1 en $(-1, 2)$. Sea $\gamma(t) = g \circ \vec{\alpha}(t)$, con $\vec{\alpha}(t) = (-\cos(t), e^t)$, $t \in \mathbb{R}$.

a) Calcule $\nabla g(-1, 1)$

b) Calcule aproximadamente $\gamma(-0.05)$ mediante una aproximación lineal.

$$f \in C^1 \Rightarrow f \text{ dif}$$

$$\bar{h}(x, y) = (xy, x^2 + y^2)$$

dif por comp.
polinómicas.

$$g(x, y) = f(\bar{h}(x, y)) \text{ dif.}$$

$$Dg(-1, 1) = Df(\bar{h}(-1, 1)) \cdot D\bar{h}(-1, 1)$$

$$Dg(-1, 1) = Df(-1, 2) \cdot D\bar{h}(-1, 1)$$

$$\bar{h}(x, y) = (xy, x^2 + y^2)$$

$$D\bar{h} = \begin{bmatrix} y & x \\ 2x & 2y \end{bmatrix} \quad D\bar{h}(-1, 1) = \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}$$

$$P(r, s) = 1 - r + s$$

$$P'_r(-1, 2) = -1 = f'_r(-1, 2) \quad DF(-1, 2) = \begin{bmatrix} -1 & 1 \end{bmatrix}$$

$$P'_s(-1, 2) = 1 = f'_s(-1, 2)$$

$$Dg(-1, 1) = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}$$

$$Dg(-1, 1) = \begin{bmatrix} -3 & 3 \end{bmatrix}$$

$$\nabla g(-1, 1) = (-3, 3)$$

Sea $\gamma(t) = g \circ \vec{\alpha}(t)$, con $\vec{\alpha}(t) = (-\cos(t), e^t)$,

$$\gamma(0) = g(\vec{\alpha}(0)) \quad D\gamma(0) = Dg(\vec{\alpha}(0)) \cdot D\vec{\alpha}(0)$$

$$\gamma(0) = g(-1, 1) = f(-1, 2) \quad D\gamma(0) = Dg(-1, 1) \cdot D\vec{\alpha}(0)$$

$$\gamma(0) = p(-1, 2) \quad D\gamma(0) = \begin{bmatrix} -3 & 3 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\gamma(0) = 4 \quad D\gamma(0) = 3 = \gamma'(0)$$

$$D\vec{\alpha} = \begin{bmatrix} \sin t \\ e^t \end{bmatrix}$$

$$D\vec{\alpha}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\gamma(t) \approx \gamma(0) + \gamma'(0) \cdot t$$

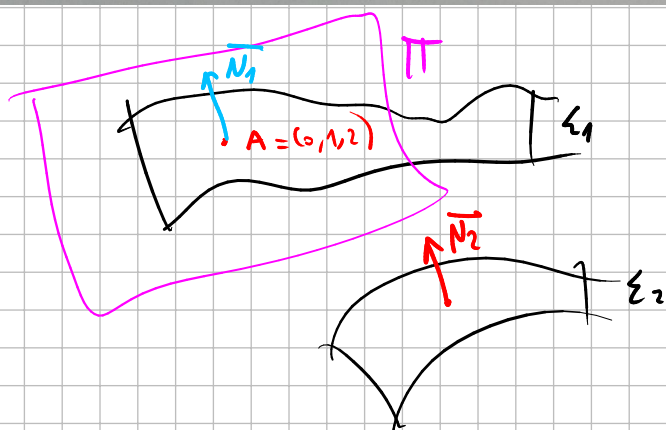
$$\gamma(-0,05) \approx 4 + 3 \cdot (-0,05) = 3,85$$

$$\gamma(-0,05) \approx 3,85$$

■ Ejercicio 5. Sean Σ_1 la superficie de ecuación vectorial

$$\vec{X} = (u + v, u^2, v - u), \quad (u, v) \in \mathbb{R}^2$$

y $\Sigma_2 : x^2 + y^2 + z^2 = 3$. Sea Π el plano tangente a Σ_1 en $A = (0, 1, 2)$. Halle los puntos $P \in \Sigma_2$ para los cuales el plano tangente a Σ_2 en P es paralelo a Π .



$$(u_0 + v_0, u_0^2, v_0 - u_0) = (0, 1, 2)$$

$$\begin{aligned} u_0 + v_0 &= 0 \\ + \quad v_0 - u_0 &= 2 \\ \hline 2v_0 &= 2 \\ v_0 &= 1 \end{aligned}$$

$$\begin{aligned} u_0 + 1 &= 0 \\ u_0 &= -1 \end{aligned}$$

$$(u_0, v_0) = (-1, 1)$$

$$\vec{X} = (u + v, u^2, v - u),$$

$$\vec{X}'_u = (1, 2u, -1) \xrightarrow{(-1, 1)} (1, -2, -1)$$

$$\vec{X}'_v = (1, 0, 1) = (1, 0, 1)$$

$$(-2, -2, 2) = \vec{N}_1 = (-1, -1, 1)$$

es lo mismo usar

$$\vec{N}_2 = k \vec{N}_1 = (-k, -k, k)$$

$\hookrightarrow k \neq 0$

$$F = 3$$

$$x^2 + y^2 + z^2 = 3$$

$$F(x, y, z)$$

$$\nabla F \perp F = k$$

$$\nabla F = (2x, 2y, 2z) = (-k, -k, k)$$

$$(x, y, z) = \left(-\frac{k}{2}, -\frac{k}{2}, \frac{k}{2}\right)$$

$$\frac{k^2}{4} + \frac{k^2}{4} + \frac{k^2}{4} = 3$$

$$3k^2 = 12 \quad k^2 = 4 \quad k = \pm 2$$

$$(x, y, z) = \begin{cases} (-1, -1, 1) \\ (1, 1, -1) \end{cases} \quad \text{P.}$$